

Sample Paper 2025-26
Class : IX
Subject : Artificial Intelligence (Code-417)

Duration : 2 hours

M.M. : 50

General Instructions :

1. Please read the instructions carefully.
2. This Question paper consists of 21 questions in **two** sections : **Section-A & Section-B**.
3. **Section-A** has objective type questions whereas **Section-B** contains subjective type questions.
4. Out of the given **(5+16)=21** questions, a candidate has to answer **(5+10)=15** questions in the allotted (maximum) time of **2** hours.
5. All questions of a particular section must be attempted in the correct order.
6. **Section-A : Objective type questions (24 marks)**
 - (a) This section has 5 questions.
 - (b) There is no negative marking.
 - (c) Do as per the instructions given.
 - (d) Marks allotted are mentioned against each question/part.
7. **Section-B : Subjective type questions (26 marks)**
 - (a) This section has 16 questions.
 - (b) A candidate has to do 10 questions.
 - (c) Do as per the instructions given.
 - (d) Marks allotted are mentioned against each question/part.

Section – A
(Objective Type Questions)

1. Answer any 4 out of the 6 questions on Employability Skills :

(4 × 1 = 4)

- (i) Which of the following is a barrier to effective communication?
 - (a) Active listening
 - (b) Clear pronunciation
 - (c) Noise
 - (d) Proper feedback
- (ii) Self-motivation helps individuals to _____.
 - (a) Delay tasks
 - (b) Set and achieve goals
 - (c) Avoid responsibilities
 - (d) Increase distractions
- (iii) Which unit stores the data and instructions temporarily?
 - (a) Input unit
 - (b) ALU
 - (c) RAM
 - (d) ROM
- (iv) An entrepreneur who starts a new business by taking risks is known as a _____.
 - (a) Manager
 - (b) Supervisor
 - (c) Innovator
 - (d) Risk-taker
- (v) Biodiversity conservation is related to _____.
 - (a) Green skills
 - (b) Financial literacy
 - (c) Digital literacy
 - (d) Data analysis
- (vi) A person's attitude, beliefs, and values form their _____.
 - (a) Cultural background
 - (b) Emotional state
 - (c) Family structure
 - (d) Work environment

2. Answer any 5 out of the 6 questions :**(5 × 1 = 5)**

- (i) A loop that repeats until a specific condition becomes false is called a _____.
(a) Conditional loop (b) Infinite loop
(c) Definite loop (d) Controlled loop
- (ii) Data that has been organized so that it is meaningful is known as _____.
(a) Raw data (b) Information
(c) Primary data (d) Secondary data
- (iii) Which of the following represents data security?
(a) Making data readable to everyone (b) Protecting data from unauthorized access
(c) Selling personal information (d) Sharing passwords
- (iv) The main purpose of data visualization is to _____.
(a) Delete data (b) Present data in graphical form
(c) Hide complex data (d) Increase storage
- (v) Which of the following is an example of data classification?
(a) Sorting a list of students by their grades
(b) Encrypting a message for privacy
(c) Calculating the average age of a group of people
(d) Storing data in the cloud
- (vi) A probability of 0.2 indicates that the event is _____.
(a) Certain (b) Impossible (c) Unlikely (d) Likely

3. Answer any 5 out of the 6 questions :**(5 × 1 = 5)**

- (i) Assertion (A): AI systems need large datasets for accurate predictions.
Reason (R): Larger datasets help the model identify patterns more effectively.
Choose the correct option:
(a) Both A & R true, R explains A
(b) Both A & R true, R does not explain A
(c) A true, R false
(d) A false, R true
- (ii) What impact does data bias have on AI models?
(a) It improves the model's accuracy for all data
(b) It can lead to unfair or inaccurate predictions
(c) It speeds up the training process
(d) It makes the model more flexible
- (iii) How does AI contribute to IoT?
(a) By storing data in a database
(b) By analyzing and making decisions based on data collected by IoT devices
(c) By providing a platform to store large amounts of data
(d) By creating wireless connections for IoT devices

(iv) Which of the following is an ethical challenge of AI?

- (a) Faster computing
- (b) Privacy concerns
- (c) Better accuracy
- (d) Reduced workload

(v) Why is it important to define the "When" in the Problem Canvas?

- (a) It helps to set the budget for the project
- (b) It determines the timeline and urgency of solving the problem
- (c) It helps identify who is responsible for solving the problem
- (d) It gives an overview of the stakeholders involved

(vi) Which mathematical concept is crucial for training models in machine learning, especially for optimization?

- (a) Probability distributions
- (b) Derivatives and gradients
- (c) Set theory
- (d) Linear equations

4. Answer any 5 out of the 6 questions :

(5 × 1 = 5)

(i) In a GAN, which two models work together?

- (a) Encoder and Decoder
- (b) Generator and Discriminator
- (c) Classifier and Regressor
- (d) Input and Output

(ii) What will be the output?

`L = [5, 10, 15, 20]`

`L.append(25)`

`print(L)`

- (a) [5,10,15]
- (b) [25,20,15,10,5]
- (c) [5,10,15,20,25]
- (d) Error

(iii) An event that is completely certain has probability equals to _____ .

- (a) 0
- (b) 1
- (c) Between 0 and 1
- (d) Greater than 1

(iv) What will be the output of :

`a = 10`

`b = 3`

`print(a // b)`

- (a) 3.33
- (b) 3
- (c) 0
- (d) 1

(v) Which is the first stage of the AI Project Cycle?

- (a) Modelling
- (b) Evaluation
- (c) Problem Scoping
- (d) Data Acquisition

(vi) A box contains 4 green and 6 yellow balls. Probability of picking a green ball is:

- (a) 4/10
- (b) 6/10
- (c) 1/4
- (d) 2/3

5. Answer any 5 out of the 6 questions :

(5 × 1 = 5)

(i) What is the primary purpose of a Variational Autoencoder (VAE)?

- (a) To predict future values based on data
- (b) To classify images into categories
- (c) To generate new data that is similar to the original data
- (d) To reduce the size of the dataset

(ii) What does the pop() method do when used on a list in Python?

- (a) Adds an element to the beginning of the list
- (b) Removes and returns the last element of the list
- (c) Inserts an element at a specific position
- (d) Sorts the list in ascending order

(iii) What is the main purpose of a regression model in machine learning?

- (a) To classify data into categories
- (b) To predict a continuous value based on input data
- (c) To visualize data in a graphical form
- (d) To generate fake data

(iv) A bar chart is useful for representing _____.

- | | |
|----------------------|--------------------|
| (a) Trends over time | (b) Proportions |
| (c) Comparisons | (d) Large datasets |

(v) In Generative AI, what does the term “training data” refer to?

- (a) Data that is used for testing the AI system
- (b) Data the AI uses to learn patterns and create new content
- (c) Data that is used to monitor performance
- (d) Data used for evaluating errors in predictions

(vi) Find the output of the following code :

```
for i in range(10, 4, -2):  
    print(i, end=" ")
```

- | | | | |
|------------|--------------|--------------|----------|
| (a) 10 8 6 | (b) 10 8 6 4 | (c) 10 9 8 7 | (d) 10 6 |
|------------|--------------|--------------|----------|

SECTION – B

(Subjective Type Questions)

Answer any 3 out of the given 5 questions on Employability Skills in 20–30 words. (3 × 2 = 6)

6. Mention any two strategies to improve time management.
7. List any four principles of sustainable development.
8. Define cultural sensitivity with an example.
9. Differentiate between wired and wireless connection.
10. State any two advantages of running a corporation form of business organization.

Answer any 4 out of the 6 questions in 20–30 words.

(4 × 2 = 8)

11. Write any two disadvantages of AI automation.
12. Define features and labels in machine learning.
13. What is a chatbot? Give an example.
14. State any two benefits of data visualization.
15. What is the purpose of problem scoping?
16. Differentiate between quantitative and qualitative data.

Answer any 3 out of the given 5 questions in 50–80 words .

(3 × 4 = 12)

17. A hospital wants to create an AI system that can analyse patients' symptoms and suggest whether they should visit a doctor immediately or follow basic home care. The AI will use previous patient records and symptom descriptions to make predictions. The developers want the system to be safe, fair, and easy to use.

Based on the above case, answer the following:

- (i) Which *two stages* of the **AI Project Cycle** will be most important at the beginning of this project, and why? [2]
- (ii) State *two AI Ethics principles* the developers must follow while creating this system. [1]
- (iii) How can the developers use **AI Reflection** after the system is deployed? [1]

18. A grocery store wants to use AI to improve its business. They have a dataset of customers' past purchases. The store has two goals:

Predict what items a customer is likely to buy next.

Group customers into categories based on their shopping patterns.

Identify which goal should use supervised learning and which should use unsupervised learning. Explain your reasoning for both. [2+2]

19. Tiya, a council member in her school, wants to determine the most popular subject among students : Math, Science or English. Which type of tool should she use to display this information effectively? How would it help to understand the data easily? [2+2]

20. What are logical operators? State the difference between division and modulus with an example of each. [1+3]

21. Explain any two applications of statistics in AI with suitable examples. [2+2]
